

Technology Stack

1. System Technology Layout

CreditGuard AI leverages modern frameworks to maintain a stateless, lightweight deployment suitable for serverless platforms:

Layer	Technology Option	Reasoning / Choice
Frontend UI	HTML5, Vanilla CSS, JS	Maximum flexibility, smooth animations, no heavy frameworks
Application Engine	Flask 3.0.x	Lightweight Python WSGI controller suited for API routes
Storage Persistence	SQLite 3.x / /tmp	Dynamic serverless database logs, redirected to writable directories
Modeling Libraries	Scikit-Learn, Pandas	Advanced ML preprocessing pipelines and Logistic Regression
Hosting Environment	Vercel Serverless	High availability, zero-maintenance global scaling